
Reinforcement Learning, Assignment 4

Rwik Kamilya (s4822285)

Abstract

This report implements Proximal Policy Optimization with clipping (PPO-Clip) from scratch on CartPole-v1 and studies which mechanisms explain its sample efficiency and stability. PPO extends Advantage Actor-Critic by reusing on-policy rollouts across minibatch epochs while limiting policy drift through a clipped probability-ratio objective. Under a 10^6 step budget, default PPO, tuned PPO and ablations of clipping, GAE, advantage normalization and repeated update epochs are evaluated. Tuned PPO gives the strongest early-learning PPO result, with $AULC_{200k} = 0.587$, $t_{475} = 126362 \pm 23383$, and final return 500.0 ± 0.0 . Removing clipping still solves but causes severe post-solve instability, while disabling advantage normalization gives the strongest PPO retention. Contextual DQN, REINFORCE, AC, and A2C baselines are included for scale, but are not protocol-matched.

1. Introduction

This project follows a standard progression from the previous assignments in deep reinforcement learning. In initial assignments, tabular methods introduced value estimation, exploration, and backup targets. Then, DQN replaced the table by a neural approximation of $Q(s, a)$, making value-based learning applicable to continuous observations but introducing instability from bootstrapping, correlated data, and shared parameters (Mnih et al., 2015). REINFORCE, actor-critic (AC), and Advantage Actor-Critic (A2C) shifted the focus from greedy value-based action selection to direct policy optimization, using a learned critic or advantage estimate to reduce policy-gradient variance (Williams, 1992; Sutton & Barto, 2018).

The basic actor-critic update is fragile, as large policy-gradient steps can make the updated policy differ substantially from the behaviour policies that generated the sampled trajectories. This mismatch can become problematic when an on-policy rollout is reused for multiple gradient steps. While data reuse improves sample efficiency, repeated updates can over-optimize a stale policy-gradient objective. PPO addresses this issue by retaining the actor-critic struc-

ture while replacing basic advantage-weighted objective with a clipped surrogate objective (Schulman et al., 2017). The clipping term discourages updates that move the new policy too far from the rollout policy, providing a simple first-order alternative to explicit trust-region optimization.

This report studies PPO-Clip on CartPole-v1. The research question we explore is: *which PPO mechanisms matter most for sample efficiency and robustness: clipping, GAE, advantage normalization, or repeated minibatch epochs?* This question is addressed by a from-scratch PPO implementation, controlled ablations, training diagnostics, and a tuned configuration selected from the observed mechanism-level behaviour. In the Cartpole setting, we investigate why different PPO variants learn, collapse, or retain a solved policy.

2. Theory

Consider an episodic Markov decision process with state $s_t \in \mathcal{S}$, action $a_t \in \mathcal{A}$, reward $r_t \in \mathbb{R}$, discount factor $\gamma \in [0, 1]$, stochastic actor $\pi_\theta(a_t|s_t)$, and critic $V_\phi(s_t)$. The actor parameters θ define a categorical distribution over the two CartPole actions, while the critic parameters ϕ define a scalar state-value estimate. While, actor and critic heads share a two-layer neural body, they optimize separate policy and value losses.

The A2C policy-gradient loss increases the log-probability of sampled actions whose estimated advantage $\hat{A}_t$ is positive and decreases it when $\hat{A}_t$ is negative:

$$L_\pi^{\text{A2C}}(\theta) = -\mathbb{E}_t[\log \pi_\theta(a_t|s_t) \hat{A}_t]. \quad (1)$$

This objective assumes that the policy being optimized is close to the policy that produced the data. In PPO, this dependence is made explicit by comparing the current policy π_θ with the rollout policy $\pi_{\theta_{\text{old}}}$. The probability ratio is

$$\rho_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)} = \exp(\log \pi_\theta(a_t|s_t) - \log \pi_{\theta_{\text{old}}}(a_t|s_t)). \quad (2)$$

The clipped PPO actor objective is minimized as

$$L_\pi^{\text{clip}}(\theta) = -\mathbb{E}_t \left[\min \left(\rho_t(\theta) \hat{A}_t, \text{clip}(\rho_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right], \quad (3)$$

where $\epsilon > 0$ is the clipping coefficient. If $\hat{A}_t > 0$, the clipped objective prevents the probability of a good sampled

action from being increased beyond the factor $1 + \epsilon$. If $\hat{A}_t < 0$, it prevents the probability of a bad sampled action from being decreased beyond $1 - \epsilon$. Thus, the surrogate is flattened exactly when the update would improve the objective mainly by moving too far from $\pi_{\theta_{\text{old}}}$. The no-clipping ablation replaces this loss by $-\mathbb{E}_t[\rho_t(\theta)\hat{A}_t]$.

The critic is trained by regression to return targets $\hat{R}_t$, and the complete minimized objective is

$$\begin{aligned} L^V(\phi) &= \frac{1}{2} \mathbb{E}_t[(\hat{R}_t - V_\phi(s_t))^2], \\ L(\theta, \phi) &= L_\pi^{\text{clip}}(\theta) + c_v L^V(\phi) - c_e \mathbb{E}_t[\mathcal{H}(\pi_\theta(\cdot|s_t))]. \end{aligned} \quad (4)$$

Here c_v is the value-loss coefficient, c_e is the entropy coefficient, and $\mathcal{H}$ is policy entropy. Entropy regularization discourages premature collapse to a deterministic policy.

Advantages are computed using generalized advantage estimation (GAE) (Schulman et al., 2016). With terminal mask $d_t \in \{0, 1\}$,

$$\begin{aligned} \delta_t &= r_t + \gamma(1 - d_t)V_\phi(s_{t+1}) - V_\phi(s_t), \\ \hat{A}_t^{\text{GAE}(\gamma, \lambda)} &= \sum_{l=0}^{\infty} (\gamma\lambda)^l \delta_{t+l}. \end{aligned} \quad (5)$$

The parameter $\lambda \in [0, 1]$ balances bias and variance. $\lambda = 0$ reduces the estimate to one-step TD residual information, while larger values include longer-horizon residuals. Return targets are then $\hat{R}_t = \hat{A}_t + V_\phi(s_t)$.

Relative to A2C, PPO introduces three changes. First, the actor loss uses the probability ratio $\rho_t(\theta)$ rather than only $\log \pi_\theta(a_t|s_t)$. Second, the same rollout is reused over several shuffled minibatch epochs. Third, this reuse is stabilized through clipping, GAE, advantage normalization, entropy regularization, gradient clipping, and diagnostic monitoring of approximate KL and clip fraction.

Algorithm 1 Implemented PPO-Clip

- 1: Initialize shared actor-critic network and rollout buffer.
 - 2: Collect T transitions using $\pi_{\theta_{\text{old}}}$; store rewards, values, actions, and old log-probabilities.
 - 3: Bootstrap through truncations and compute GAE advantages backwards.
 - 4: Form return targets $\hat{R}_t = \hat{A}_t + V_\phi(s_t)$.
 - 5: Shuffle the rollout and repeat minibatch optimization for K epochs.
 - 6: Recompute log-probabilities, entropy, values, ratios, and clipped actor loss.
 - 7: Minimize actor loss + value loss – entropy bonus with gradient clipping.
 - 8: Log evaluation return, approximate KL, clip fraction, entropy, and critic loss.
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3. Experiments

3.1. Setup

In the CartPole-v1 (Towers et al., 2024) setting, the observation vector contains cart position, cart velocity, pole angle, and pole angular velocity. The action space contains two discrete actions. Each non-terminal step gives reward +1, and the maximum episode return is 500. Deterministic evaluation is performed every 10,000 environment steps over 20 greedy episodes.

The default network is a shared two-layer multilayer perceptron with 64 hidden units, Tanh activations, orthogonal initialization, a categorical actor head, and a scalar critic head. Default PPO uses $\gamma = 0.99$, $\lambda = 0.95$, learning rate $3 \cdot 10^{-4}$, rollout length 2048, minibatch size 256, 10 update epochs, clip coefficient $\epsilon = 0.2$, entropy coefficient 0.01, value-loss coefficient 0.5, and gradient clipping at norm 0.5. Training uses a budget of 10^6 environment steps.

Six PPO variants are evaluated. PPO Full is the default configuration. PPO Tuned uses rollout length 512, minibatch size 64, 4 update epochs, entropy coefficient 0.001, and disabled advantage normalization. Four ablations isolate individual mechanisms: no clipping, $\lambda = 0$, no advantage normalization, and a single update epoch. PPO Full and PPO Tuned use 5 seeds; ablations use 3 seeds. All PPO metrics are computed from raw deterministic evaluation logs.

The main metrics are area under the learning curve over the first 200,000 environment steps, AULC_{200k} , first solve time t_{475} , solve rate, final evaluation return, post-solve worst return, and retention. AULC_{200k} is normalized by the maximum possible CartPole return, so higher values indicate better early sample efficiency. t_{475} is the first evaluation checkpoint at which mean deterministic return reaches 475. Post-solve worst return is the minimum evaluation return after first reaching 475, and retention is the fraction of post-solve checkpoints with return at least 475.

The implementation explicitly uses clipped policy ratios, minibatch stochastic gradient descent, multiple epochs over each rollout, GAE, advantage normalization unless ablated, entropy regularization, gradient clipping, approximate-KL logging, clip-fraction logging, seed aggregation, and correct Gymnasium termination/truncation handling. It does not use SAC, replay buffers, target networks, reward normalization, observation normalization, vectorized environments, learning-rate schedules, target-KL early stopping, or model checkpointing.

Saved Assignment 2/3 results for DQN, REINFORCE, AC, and A2C are included as contextual baselines. These traces are useful for scale, but they are not strictly protocol-matched: PPO uses deterministic evaluation checkpoints,

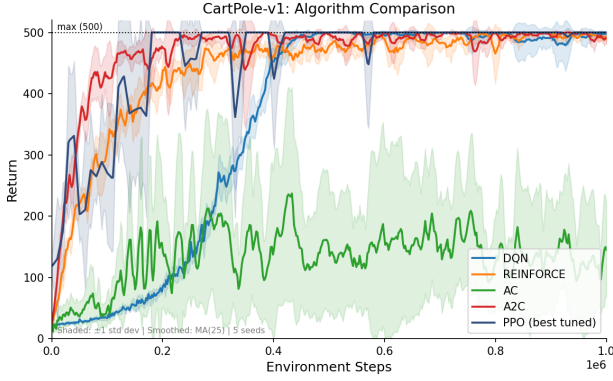


Figure 1: Contextual CartPole-v1 comparison. PPO uses deterministic evaluation every 10,000 steps over 20 episodes; DQN, REINFORCE, AC, and A2C are reconstructed from previous-assignment training traces, so they are scale baselines rather than protocol-matched comparisons. Lines show seed means and shaded regions show ± 1 standard deviation.

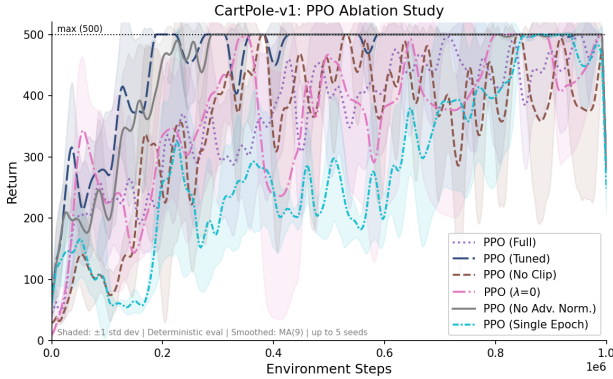


Figure 2: Controlled PPO ablations on CartPole-v1. Deterministic evaluation is performed every 10,000 steps over 20 greedy episodes. Curves show seed means with ± 1 standard deviation and isolate clipping, GAE depth, advantage normalization, and repeated minibatch epochs.

whereas previous baselines are reconstructed from saved training-return traces.

3.2. Results

Table 1 shows that every PPO variant reaches the solve threshold in all seeds, but the path to solving differs substantially. PPO Tuned is the strongest early-learning PPO variant, with $\text{AULC}_{200k} = 0.587$ and $t_{475} = 126362 \pm 23383$. PPO Full also reaches final return 500.0 ± 0.0 , but has lower early-learning area and weaker post-solve retention. Thus, final evaluation alone is insufficient: the same final return can hide large differences in sample efficiency and stability.

The ablations identify the main mechanisms. The single-epoch variant is the weakest PPO configuration by sample efficiency, with $\text{AULC}_{200k} = 0.193$ and $t_{475} = 527701 \pm$

253392. This confirms that repeated minibatch updates are central to PPO’s practical data efficiency. Without rollout reuse, the method behaves much closer to a slow on-policy actor-critic learner.

The no-clipping ablation is the clearest negative control. It eventually reaches final return 500.0 ± 0.0 , but its post-solve worst return is only 12.5 ± 1.0 . This indicates severe policy instability after first solving. The result is consistent with PPO’s motivation: once the clipped ratio constraint is removed, repeated epochs can move the policy too far from the rollout policy, making performance retention unreliable.

The $\lambda = 0$ ablation also reaches final return 500.0 ± 0.0 , but has lower retention and weaker post-solve stability than PPO Tuned and PPO No Adv. Norm. One-step TD residuals are sufficient to solve dense-reward CartPole, but the large variability in t_{475} and low post-solve worst return suggest that longer-horizon GAE still provides a more reliable improvement signal.

The no-advantage-normalization ablation is the strongest stability result among PPO variants. Although its $\text{AULC}_{200k} = 0.493$ is lower than PPO Tuned, it has the best PPO post-solve worst return, 278.9 ± 156.4 , and equal retention of 96%. This should be interpreted as an environment-specific result rather than a general PPO recommendation. In dense-reward CartPole, preserving the raw magnitude of advantages can provide useful update-scale information; in higher-variance or sparse-reward tasks, normalization may still improve conditioning.

A2C has the highest contextual AULC_{200k} and fastest contextual t_{475} among all displayed methods. This does not imply a strict algorithmic dominance over PPO, because the previous baselines are logged as training-return traces rather than deterministic evaluation checkpoints. The comparison nevertheless shows that CartPole is simple enough for A2C to be highly competitive, and that PPO’s value in this report lies less in outperforming A2C and more in exposing the mechanisms that permit controlled on-policy data reuse.

Diagnostic logs support the ablation interpretation. PPO No Clip shows the largest policy-ratio excursions, matching its very low post-solve worst return. PPO Single Epoch keeps entropy high for longer and learns slowly, matching its delayed solve time. PPO Tuned reduces entropy while keeping policy drift controlled, explaining its stronger early-learning area and retention. Thus, identical final returns hide distinct KL, ratio, entropy, and critic-loss dynamics.

Table 1: Performance on CartPole-v1 under a 10^6 environment-step budget. PPO variants form the controlled comparison; DQN, REINFORCE, AC, and A2C are contextual previous-assignment baselines. $AULC_{200k}$ measures normalized early-learning area under the curve; t_{475} is the first deterministic evaluation checkpoint with mean return at least 475. Post-worst is the minimum deterministic evaluation return after first solving; retention is the fraction of post-solve checkpoints with return at least 475.

Method	n	$AULC_{200k} \uparrow$	$t_{475} \downarrow$	Solve $\uparrow$	Final eval $\uparrow$	Post-worst $\uparrow$	Retention $\uparrow$
PPO Full	5	0.373	136806 ± 50985	100%	500.0 ± 0.0	134.7 ± 10.6	62%
PPO Tuned	5	0.587	126362 ± 23383	100%	500.0 ± 0.0	181.3 ± 55.5	96%
PPO No Clip	3	0.268	174763 ± 11863	100%	500.0 ± 0.0	12.5 ± 1.0	73%
PPO $\lambda = 0$	3	0.389	164523 ± 114607	100%	500.0 ± 0.0	68.2 ± 45.9	66%
PPO No Adv. Norm.	3	0.493	144043 ± 25923	100%	500.0 ± 0.0	278.9 ± 156.4	96%
PPO Single Epoch	3	0.193	527701 ± 253392	100%	425.1 ± 106.0	192.5 ± 59.6	63%
A2C [†]	5	0.797	101204 ± 20470	100%	492.0 ± 5.2	324.8 ± 53.2	83%
REINFORCE [†]	5	0.623	266654 ± 24617	100%	494.4 ± 1.8	399.7 ± 25.0	70%
AC [†]	5	0.186	412956 ± 183552	40%	122.8 ± 66.6	57.0 ± 11.4	0%
DQN [†]	5	0.103	409637 ± 17643	100%	495.9 ± 8.0	449.5 ± 30.2	92%

[†]Contextual baselines from saved training-return traces, not deterministic PPO-style evaluation logs.

4. Discussion

The main empirical lesson is that PPO’s behaviour is determined by the interaction between its clipped surrogate and its engineering choices. Clipping controls policy drift, but clipping alone does not guarantee fast learning. Repeated minibatch epochs provide the data reuse that separates PPO from a simpler on-policy actor-critic method. GAE shapes the bias–variance profile of the advantage estimate. Entropy regularization controls the exploration–exploitation transition. Advantage normalization changes the scale information available to the actor update.

PPO Tuned is the best sample-efficient PPO variant in this study. Its improvement over PPO Full is substantial: $AULC_{200k}$ increases from 0.373 to 0.587, while t_{475} decreases from 136806 to 126362 steps. Shorter rollouts and smaller minibatches allow more frequent policy updates; fewer epochs reduce excessive reuse of each rollout; lower entropy regularization allows sharper exploitation once balancing behaviour is discovered; and disabling advantage normalization preserves reward-scale information that appears useful in this dense-reward environment.

The strongest robustness result is not identical to the strongest early-learning result. PPO No Adv. Norm. has lower early-learning area than PPO Tuned, but the best PPO post-solve worst return. This difference shows why multiple metrics are needed. $AULC_{200k}$ measures early learning, t_{475} measures first solve time, final return measures end-of-training performance, and post-solve worst return measures whether the solved policy is retained. A single metric would miss this trade-off.

Several limitations remain. First, the previous DQN, REINFORCE, AC, and A2C baselines are not protocol-matched to PPO because they use saved training-return traces rather

than deterministic evaluation logs. Second, ablations use 3 seeds, so stability conclusions should be treated as suggestive rather than definitive. Third, CartPole-v1 is low-dimensional and dense-reward; the advantage-normalization result may not transfer to sparse-reward or continuous-control tasks. Fourth, the implementation omits optional PPO refinements such as learning-rate annealing, target-KL early stopping, vectorized environments, explained-variance diagnostics, and checkpoint selection. These omissions keep the implementation transparent, but limit claims about state-of-the-art PPO performance.

5. Conclusion

A from-scratch PPO-Clip implementation was evaluated on CartPole-v1 through controlled ablations, diagnostics, and contextual comparison to earlier DQN, REINFORCE, AC, and A2C results. PPO satisfies the actor-critic structure studied previously while replacing the plain A2C policy update with a clipped probability-ratio objective that enables safer rollout reuse. All PPO variants solve CartPole, but their learning dynamics differ sharply. PPO Tuned gives the strongest early-learning performance, PPO No Adv. Norm. gives the strongest post-solve stability among PPO variants, PPO No Clip exposes the importance of policy-drift control, and PPO Single Epoch confirms the importance of repeated minibatch updates. Future work should rerun all previous baselines under the same deterministic evaluation protocol, increase ablation seed counts, and test whether the advantage-normalization and entropy findings persist in harder sparse-reward or continuous-control environments.

References

- Mnih, V., Kavukcuoglu, K., Silver, D., Rusu, A. A., Veness, J., Bellemare, M. G., Graves, A., Riedmiller, M., Fidjeland, A. K., Ostrovski, G., et al. Human-level control through deep reinforcement learning. *Nature*, 518(7540): 529–533, 2015.
- Schulman, J., Moritz, P., Levine, S., Jordan, M., and Abbeel, P. High-dimensional continuous control using generalized advantage estimation. In *International Conference on Learning Representations*, 2016.
- Schulman, J., Wolski, F., Dhariwal, P., Radford, A., and Klimov, O. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.
- Sutton, R. S. and Barto, A. G. *Reinforcement Learning: An Introduction*. MIT Press, 2 edition, 2018.
- Towers, M., Kwiatkowski, A., Terry, J. K., Balis, J. U., De Cola, G., Deleu, T., Goulão, M., Kallinteris, A., Krimmel, M., KG, A., et al. Gymnasium: A standard interface for reinforcement learning environments. *arXiv preprint arXiv:2407.17032*, 2024.
- Williams, R. J. Simple statistical gradient-following algorithms for connectionist reinforcement learning. *Machine Learning*, 8(3):229–256, 1992.